

INTERACTIVE INFORMATION VISUALIZATION

VIZIONARIES TEAM

To what extent does the development of cycling infrastructure help reduce the number and severity of cycling accidents?

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Section 1: List of Dataset

0.0.1 Find by HOUSSENBAY:

BAAC (Bulletin d'Analyse des Accidents Corporels)

- **Source:** French Ministry of Interior
- **Access:** <https://www.data.gouv.fr/fr/datasets/bases-de-donnees-annuelles-des-accidents-corporels-de-la-circulation-routiere-annees-de-2005-a-2023>
- **Description:** Official French traffic accident database with detailed information about all reported accidents, including location, severity, and circumstances.
- **Update Frequency:** Annual
- **First date available:** 2005
- **Last date available:** 2023
- **Format:** CSV files with separate tables for characteristics, locations, vehicles, and casualties all the information about the column can be found on this document

0.0.2 Find by BOUSALEM:

Paris Traffic Data

- **Source:** Paris Open Data Portal
- **Access:** <https://opendata.paris.fr/explore/dataset/comptage-velo-donnees-compteurs/>
- **Description:** Bicycle traffic counts from automated counters throughout Paris
- **Update Frequency:** Hourly
- **Format:** CSV or API access

We can also add some weather data from Météo-France for example to understand more precisely the condition of the accident.

- **Source:** Météo-France

- **Access:** <https://donneespubliques.meteofrance.fr/>
- **Description:** Historical weather conditions that might affect cycling safety
- **Update Frequency:** Daily
- **Format:** CSV or API access

Section 2: Selected Dataset

Our main dataset will be the BAAC dataset, for one simple reason the data is collected by French law enforcement and transportation authorities, ensuring official documentation of all reported incidents. Each accident record includes detailed information about location, time, severity, and contributing factors. We have chosen this three dataset in particular first because of their reliability.

Our goal is to enhance this dataset with other official datasets, such as the Paris Traffic Data, to better understand cyclist traffic for specific routes or infrastructure, as well as the impact of weather on the infrastructure chosen by cyclists under different conditions. Additionally, we aim to analyze the evolution of accidents based on the creation of new infrastructure. For this purpose, we are not relying on a single dataset from a specific date but rather on multiple datasets spanning a time range, specifically from 2005 to 2023.

Section 3: Research Question Evolution

This section is really particular for us because we misunderstood the 3 questions to give and we only gave one question. At first, we only wanted to focus on bicycle infrastructure near bicycle sharing services such as Vélib to better understand the importance of placing bicycle sharing services at propitious locations.

We were highly inspired by this video from Le Parisien published two weeks ago that was talking about the problem of poor infrastructure planning of bike sharing services on one hand, and the city of Paris insisting on putting more people on bikes https://www.youtube.com/watch?v=8MLluoeRPU8&ab_channel=LeParisien.

We have decided to change our focus a little bit. We will still focus on the city of Paris, but this time we will broaden our approach. Instead of focusing on bike sharing users only, we will focus on all bikers. We have decided to make this choice because we struggle to find data that would specially distinguish between a regular biker and a bike sharing user. Then our new question will be: How does the development and quality of cycling infrastructure impact the frequency and severity of cycling accidents across French regions?

	Identifiant du compteur	Nom du compteur	Identifiant du site de comptage	Nom du site de comptage	Comptage horaire	Date et heure de comptage	Date d'installation du site de comptage	Lien vers p
1.	100003098-101003098	106 avenue Denfert Rochereau NE-SO	100003098	106 avenue Denfert Rochereau	7	2023-12-01T04:00:00+01:00	2012-02-22	https://filer.e-tools.com/fil
2.	100003098-101003098	106 avenue Denfert Rochereau NE-SO	100003098	106 avenue Denfert Rochereau	10	2023-12-01T05:00:00+01:00	2012-02-22	https://filer.e-tools.com/fil
3.	100003098-101003098	106 avenue Denfert Rochereau NE-SO	100003098	106 avenue Denfert Rochereau	9	2023-12-01T06:00:00+01:00	2012-02-22	https://filer.e-tools.com/fil

Figure 1: Paris Open Data Portal

	File	Num_Acc	id_usager	id_vehicule	num_veh	place	catu	grav	sexe	an_nais	trajet	secu1	secu2	secu3	locp	actp	etatp
1.	usagers-2023.csv	2023000000001	203 851 184	155 680 557	A01	1	1	4	1	1978	5	2	0	-1	-1	-1	-1
2.	usagers-2023.csv	2023000000002	203 851 182	155 680 556	A01	1	1	1	2	1997	9	1	0	-1	-1	-1	-1
3.	usagers-2023.csv	2023000000002	203 851 183	155 680 556	A01	10	3	3	1	1997	9	0	-1	-1	2	3	1
4.	usagers-2023.csv	2023000000003	203 851 180	155 680 554	B01	1	1	3	1	1987	0	2	6	0	0	0	-1
5.	usagers-2023.csv	2023000000003	203 851 181	155 680 555	A01	1	1	1	2	1984	0	1	0	0	0	0	-1
6.	usagers-2023.csv	2023000000004	203 851 175	155 680 551	B01	2	2	1	2	2001	1	1	0	-1	-1	-1	-1
7.	usagers-2023.csv	2023000000004	203 851 176	155 680 551	B01	1	1	1	2	1995	1	1	0	-1	-1	-1	-1
8.	usagers-2023.csv	2023000000004	203 851 177	155 680 552	C01	1	1	1	1	1968	9	1	0	-1	-1	-1	-1
9.	usagers-2023.csv	2023000000004	203 851 178	155 680 552	C01	2	2	4	1	1956	0	1	0	-1	-1	-1	-1

Figure 2: BAAC Dataset

Section 4: Data Cleaning Process

For the BAAC dataset, we have to remove all data that does not involve a bike and other users, and we only want to focus on the Paris area. One particularity of the BAAC dataset is that it is divided into multiple files, exactly like a SQL database, so we have to merge them to retain only the useful information. On one hand, this is good because we have detailed information about accidents and many things to visualize. On the other hand, we haven't had the time to do it perfectly.

For the Paris Open Data Portal and Météo-France data, we have nothing to clean since we can focus only on the city of Paris. We have removed some non usefull information such as photo url taken by each counter.

We also encounter some memory limit when treating the Paris Open Data Portal and we will need more time to carefully link each of the dataset.

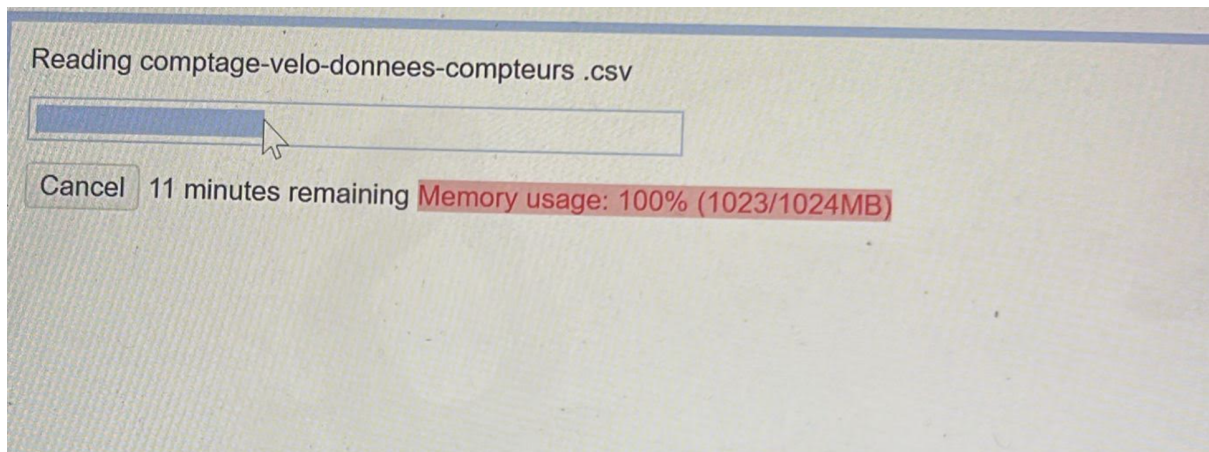


Figure 3: Memory limit

Section 5: Dataset Access

<https://drive.google.com/drive/folders/1VXob6n_qioE-uwWD7zQFaiAmeEZE3PJb?usp=sharing>